

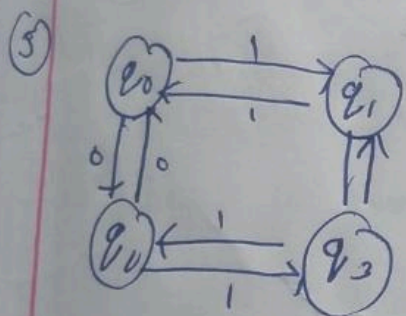
Assignment - 1

① prove $1^3 + 2^3 + \dots + n^3 = \frac{n^2(n+1)^2}{4}$

② $\sqrt{2}$ is a rational number.

③ construct a DFA that string 0,1 but the string is starting with 10.

④ construct a DFA to accept string 0's and 1's that start with '1' and ends with '1'.



check with the following string.

1) 1001

2) 101010

3) 1011

Answers

A) $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$

Assume $n=0(0,1)\dots$ then

$$2 \left(\frac{1(1+1)}{2} \right)^4 = \left(\frac{2}{2} \right)^4 = 1^4 = 1$$

LHS = RHS

if $n=k$

$1^3 + 2^3 + 3^3 + \dots + k^3 = \left(\frac{k(k+1)}{2} \right)^2 \rightarrow \textcircled{1}$

$$n = k+1$$

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \left[\frac{(k+1)(k+1+1)}{2} \right]^2$$

from ① & ② Add $(k+1)^3$ on both sides

$$1^3 + 2^3 + 3^3 + \dots + k^3 + (k+1)^3 = \left[\frac{(k+1)(k+2)}{2} \right]^2 + (k+1)^3$$

$$= \left(\frac{k(k+1)}{2} \right)^2 + (k+1)^3$$

$$= \frac{k^2(k+1)^2 + 4(k+1)^3}{4}$$

$$= \frac{(k+1)^2 (k^2 + 4k + 4)}{4}$$

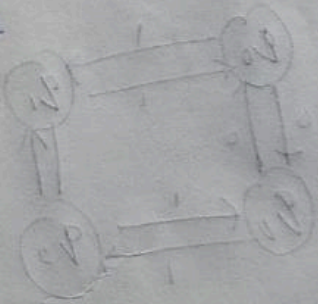
$$= \frac{(k+1)^2 (k^2 + 4k + 4)}{4}$$

$$= \frac{(k+1)^2 (k+2)^2}{4}$$

$$= \left[\frac{(k+1)(k+2)}{2} \right]^2$$

= R.H.S

∴ Hence Proved //



②A) let us assume that $\sqrt{2}$ is a rational number with a and b as co prime integer and $b \neq 0$.

$$\sqrt{2} = \frac{a}{b}$$

squaring on both sides

$$(\sqrt{2})^2 = \left(\frac{a}{b} \right)^2$$

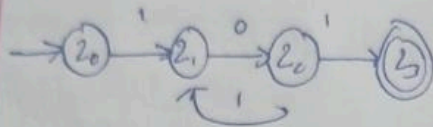
$$2 = \frac{a^2}{b^2}$$

$$b^2 \cdot 2 = a^2$$

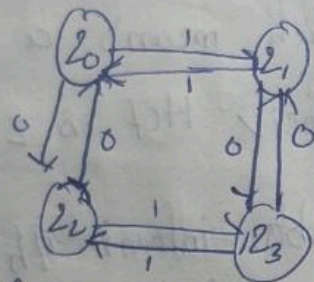
∴ Here $2a^2$ is a multiple of 2 and it is even.

(4) Given that string is 0 is and 1's
string start with 1 and ends with 1.

initial = 1 [00, 01, 10, 11]
final = 1 [11, 110111, 1101, 1100]



(5) Given DFA diagram is



To check string accepted by DFA;

0 1001 $\delta(q_0, 1001)$
 $\delta(q_1, 001)$
 $\delta(q_3, 01)$
 $\delta(q_1, 1)$
 $\delta(q_0)$

$\therefore$ String is accepted

1) 101010

Checking: $\delta(q_0, 101010)$
 $\delta(q_1, 01010)$
 $\delta(q_3, 1010)$
 $\delta(q_2, 1010)$

